

Loc Ngo

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Education

Military Technical Academy Hanoi, Vietnam
Master, Information Systems 2016
Thesis: Genetic programming for time series forecasting

Hanoi Aptech Hanoi, Vietnam
Higher Diploma, Software Engineering 2010-2012
Concentrated on software development and engineering principles.

University of Engineering and Technology, Vietnam National University Hanoi, Vietnam
Bachelor of Science, Engineering Physics 2006-2010
The main subjects focus on mathematics, quantum physics, and nanotechnology.

Certificate

Course Certificate for Machine Learning
Stanford Online 2023
This course provides a broad introduction to machine learning, data-mining, and statistical pattern recognition.

Experience

Fortna, Inc. Atlanta, GA (Remote)
Senior Software Architect Mar 2025 – Present

- Designed and developed Order Planner, a module that intelligently controls and balances order throughput volume across the distribution center.
- Architected and implemented a human operations simulation framework used to benchmark WES performance and validate algorithmic behavior under realistic DC conditions, decoupling quality assurance from physical operational dependencies.
- Designed and implemented a custom partition support solution for NATS messaging, engineering a capability not available natively to meet the platform's scalability, ordering, and fault-tolerance requirements.
- Pioneered AI adoption initiatives by defining best practices, processes, and documentation for integrating AI tooling into the software development lifecycle, measurably improving code quality and delivery velocity.
- **Tech stack:** Java, Go, Python, NATS, Microservices, PostgreSQL, Redis, Docker, Github Action, CI/CD, OpenTelemetry, Datadog

Fortna, Inc. Atlanta, GA (Remote)
Software Architect Mar 2022 – Mar 2025

- Enhanced the internal Java framework to support Java 17+, introducing an improved dependency injection mechanism via Google Guice to increase application flexibility, extensibility, and customization.
- Designed and developed a Hardware Abstraction Service capable of managing 2,000+ concurrent bi-directional TCP-based devices across a live distribution center, with horizontal scaling support to grow well beyond that.
- Designed and implemented the Task Activity Framework, a unified specification language bridging product owners and engineers — accelerating the translation of business requirements into consistent system design and implementation across the development lifecycle.
- Supported project teams in system operations, troubleshooting, and root cause analysis (RCA). This includes working with both legacy systems built on JBoss and Oracle, as well as newer platforms.
- **Tech stack:** Java, Go, JNI, NATS, Microservices, PostgreSQL, Redis, Docker, Github Action, CI/CD, Grafana Stack, OpenTelemetry

Fortna Vietnam LLC
Senior Software Engineer

Hanoi, Vietnam
Jan 2017 – Feb 2022

- Led the architectural transformation of the company's legacy Warehouse Execution System (FortnaWES) into a modern microservices-based platform, improving scalability, maintainability, and deployment agility.
- Designed and built an internal Java-based development framework that accelerated application development and enforced consistent coding standards across all services.
- Engineered a unit-sorter simulator that enabled development, testing, and performance benchmarking of sorting services without requiring physical hardware integration, maintaining full quality and performance guarantees.
- Designed and developed the Orchestration Engine, coordinating order and work sequencing across all operational areas within the distribution center.
- Architected and built a modular micro-frontend UI platform enabling parallel development across distributed teams, improving delivery speed and long-term maintainability.
- Supported project teams in system operations, troubleshooting, and root cause analysis (RCA). This includes working with both legacy systems built on JBoss and Oracle, as well as newer platforms.
- **Tech stack:** Java, Go, JNI, Microservices, ReactJS, PostgreSQL, Oracle, ActiveMQ, Redis, Docker, Github Action, CI/CD, Grafana Stack

YCTECH (Outsourcing)
Co-Founder & Lead Architect

Hanoi, Vietnam
Jan 2016 – Jan 2017

- Founded and led the technical division of an outsourcing startup, owning architectural strategy, solution design, and end-to-end technical decision-making across client engagements.
- Designed and delivered the first version of Yojee, a real-time logistics platform handling complex supply chain operations across road, air, and ocean freight networks, now-publicly listed logistics technology platform serving global freight forwarders and shippers including Maersk, CEVA Logistics, and Geodis.
- **Tech stack:** Go, Microservices, Android, PostgreSQL, Docker, Heroku (PaaS), Redis

Viettel Cyberspace Center - Viettel Group
Backend & Big-Data Developer

Hanoi, Vietnam
Feb 2015 – June 2016

- Designed and developed distributed backend systems and analytical tools for large-scale network traffic monitoring leveraging Kafka, Storm, Hadoop, and Zookeeper.
- Architected and implemented a real-time spam SMS filtering and blocking system capable of processing nearly 300,000 SMS/minute, successfully eliminating 80–90% of spam across the entire carrier network — recognized with a Bronze Award at IT World Awards 2016
- Built a high-throughput, real-time message storage using HDFS and MapReduce, engineered to sustain throughput exceeding 1,000,000 messages/second
- **Tech stack:** Java, Microservices, Kafka, MapReduce, Storm, Spark, Avro, Parquet, ZeroMQ, Redis, AeroSpike

Wala
Android Technical Lead

Hanoi, Vietnam
Feb 2012 – Jan 2015

- Developed and shipped a full-featured Android messaging application, owning end-to-end UI implementation and core feature delivery.
- Benchmarked and optimized connection management libraries, improving network reliability and reducing latency for real-time messaging.
- Engineered photo optimization pipelines to reduce bandwidth consumption and enhance media loading performance.
- Optimized UI rendering pipelines, application performance, and battery efficiency, resulting in improved end-user experience.
- **Tech stack:** Java, Android

CNC Software (Outsourcing)

Hanoi, Vietnam

Android Technical Lead

Aug 2010 – Jan 2012

- Architected and designed mobile application frameworks and communication models aligned with client requirements and technical specifications.
- Led end-to-end solution design for client-facing Android applications, translating business concepts into efficient technical implementations.
- Researched, developed, and deployed internal libraries and frameworks to streamline the development lifecycle and accelerate delivery
- **Tech stack:** Java, Android

Personal Projects

Atticus

2025 – Present

<https://staging.atticus.vn>

Atticus is an advanced AI-driven legal assistant designed specifically for Vietnamese law and legal practice.

- Designed and implemented an event-driven architecture with core-level partitioning and sharding, enabling horizontal scalability and supporting thousands of concurrent requests.
- Developed an automated data pipeline for legal document chunking, reducing manual effort by 80% and improving processing accuracy to 90%.
- Enhanced contextual retrieval mechanisms, increasing knowledge lookup accuracy to 90%.
- Refined agent instruction design to improve task focus and boost expected outcome accuracy.
- **Tech stack:** Python, NextJS, Agno (Agent Framework), PostgreSQL (pgvector, pg_search), Github Action, GitOps Pipeline, AWS (IAM, S3, Secret, RDS), Ansible, Docker

ChromaDB Client (Java)

2025

<https://github.com/locxngo/chroma-client>

An implementation of ChromaDB client in Java used to connect to a running ChromaDB server or service.

- Designed a modular Java client for ChromaDB with extensible interfaces, enabling seamless integration with multiple embedding providers.
- Abstracted embedding generation behind configurable adapters, allowing plug-and-play support for custom or third-party embedding systems.
- **Tech stack:** Java, Maven

SqlBundle

2022

<https://github.com/locxngo/sqlbundle>

A lightweight, SQL-based database migration tool that uses plain SQL files

- Developed a lightweight SQL-based database migration tool in Go to manage schema evolution using plain SQL files.
- Designed a simple and intuitive migration workflow targeting small to medium-scale projects.
- Implemented migration execution engine supporting ordered schema updates and rollback-safe deployment processes.
- **Tech stack:** Go

Vulcan

2021

<https://github.com/locxngo/vulcan>

Container-native CI/CD platform that uses Docker containers to run automated builds, tests, and deployments based on pipeline configurations defined in a YAML file.

- Developed pipeline orchestration engine in Go to parse and execute YAML-defined CI/CD configurations.
- Integrated Docker container runtime to isolate build and testing environments for reproducible pipelines.
- **Tech stack:** Go, Docker

<https://github.com/hermes-solution>

It is solution focused on simple real-time log management for on-premises Docker environments and related utilities.

- Built backend log ingestion services in Go to process and store streaming logs into ClickHouse database.
- Configured Fluentd pipeline for reliable log aggregation and forwarding from Docker containers to backend storage.
- Designed and implemented React-based UI dashboard for real-time log visualization and log exploration.
- **Tech stack:** Go, ReactJS, Clickhouse, Fluentd

Skills & Interests

Software & System Design: Microservices, Distributed system, and Event-driven architecture.

Warehouse & Supply Chain: Warehouse Operations and Warehouse Execution System

AI Agent & Agentic: Data Pipeline, Contextual Retrieval (RAG, BM25), Agent Orchestration, Prompting techniques

Programming Languages: Java, Go, Python, Shell script, Typescript

Frameworks & Platform: Agno, Micronaut, NextJS, ReactJS, Expo, Hadoop MapReduce, Spark, Flink, Storm

Databases: PostgreSQL, Redis, MySQL, MongoDB

Vector Databases: ChromaDB, PgVector, Leann, LanceDB

Message Brokers: NATS, RabbitMQ, Redis Streams, ActiveMQ, Kafka

Containerization & IaC: Docker, Ansible

Operating Systems: Ubuntu, MacOS

CI/CD & Observability CI/CD (Github Action, GitOps pipeline), Grafana Stacks, OpenTelemetry, Datadog